A universal orthogonality-transfer principle for geometric constants

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Status: candidate full structural answer, likely valid; novelty not certified. Under the literal meaning of "equivalently expressed", all finite-point unit-sphere constants admit exact isosceles-orthogonal coordinates. This gives arbitrary many-parameter extensions of L'_{YJ}, a necessary-and-sufficient criterion for other orthogonalities, a new Pythagorean-orthogonality transfer, and exact operator versions. The source questions are informal, so claims about which formulas are *natural* or *useful* remain outside the result.

1 Current source questions and version boundary

Wang, Liu, and Bao [1] ask in Problems 5.8–5.12 which geometric constants have orthogonality-based representations, whether their two-parameter L'_{YJ} construction has more-parameter analogues, whether unit-ball or unit-sphere restrictions can generally be encoded by isosceles orthogonality, whether other orthogonalities work, and how operator versions should be treated.

The current source is arXiv:2507.17122v5, revised 18 July 2026. This matters: v5 itself adds a Birkhoff–James rectification theorem and exact formulas for several constants. Those results are prior source material here, not claimed as new. The five questions below remain on printed page 22 of v5.

2 Proof intuition

There is an exact, norm-independent change of coordinates behind every isosceles formula in the source. Given two unit vectors p, q, their half-sum $u = (p+q)/2$ and half-difference $v = (p-q)/2$ satisfy $\|u+v\| = \|u-v\| = 1$; hence $u \perp_I v$. Conversely $p = u+v$ and $q = u-v$. This is a bijection, not an inequality. Therefore every function, constraint, parameter, supremum, or infimum on two sphere points transfers exactly.

For another orthogonality relation, the corresponding issue is precisely surjectivity: can every vector z be written as a longitudinal part βx plus a part y orthogonal to x? This "rectifiability" is both necessary and sufficient for universal transfer. Birkhoff–James rectifiability is proved in v5 of the source. Pythagorean orthogonality is a new example: a one-dimensional convex forward-difference argument always chooses the required β.

3 The Hadamard transfer theorem

Let X be a nonzero real normed space and define the normalized isosceles cone

$$\mathcal{I}_X := \{(u, v) \in X^2 : u \perp_I v,\ \|u+v\| = 1\}.$$

$$C_{\mathrm{NJ}}^{(p)}(X) = \sup_{x \neq 0, x \perp_B y, \beta \in \mathbb{R}} \frac{\|(1+\beta)x + y\|^p + \|(1-\beta)x - y\|^p}{2^{p-1}\left(\|x\|^p + \|\beta x + y\|^p\right)}.$$

At the end of the article, we would like to pose some open questions:

Problem 5.8. *What kinds of geometric constants can be equivalently expressed using isosceles orthogonal constants?*

Problem 5.9. *So far, we know that $C'_{\mathrm{NJ}}(X)$ is a special case of the $L'_{\mathrm{YJ}}(\tau, v, X)$ constant, and we have proposed an isosceles orthogonal equivalent form for the $L'_{\mathrm{YJ}}(\tau, v, X)$ constant with two parameters. Then, are there any constants with more parameters that are generalized forms of $L'_{\mathrm{YJ}}(\tau, v, X)$ and can also be equivalently expressed by means of isosceles orthogonality?*

Problem 5.10. *As is well known, geometric constants are almost always studied on the unit ball or unit sphere. Is the combination of a constant and the condition of isosceles orthogonality sufficient to satisfy the aforementioned conditions?*

Problem 5.11. *In this paper, we primarily employ constants associated with isosceles orthogonality and Birkhoff orthogonality to provide equivalent characterizations of other geometric constants. This naturally raises a question: can analogous equivalent representations be established for geometric constants via other types of orthogonality?*

Inspired by Beauzamy's relevant generalization work [9], Pisier extended the concepts of the modulus of convexity and the modulus of smoothness to bounded linear operators from space X to space Y in reference [45]. Therefore, we have the following question:

Problem 5.12. *How can the geometric constants of the operator version be equivalently characterized through orthogonality?*

Acknowledgment

Figure 1: Problems 5.8–5.12 in arXiv:2507.17122v5, printed page 22.

The orthogonality condition automatically also gives $\|u - v\| = 1$.

Theorem 1 (Universal two-point transfer)**.** *The map*

$$H : S_X^2 \longrightarrow \mathcal{I}_X, \qquad H(p,q) = \left(\frac{p+q}{2}, \frac{p-q}{2}\right)$$

is a bijection with inverse $H^{-1}(u,v) = (u+v, u-v)$. Consequently, for every set of parameters Θ and every extended-real function $F : \Theta \times S_X^2 \to [-\infty, \infty]$,

$$\sup_{\theta \in \Theta,\ p,q \in S_X} F(\theta, p, q) = \sup_{\theta \in \Theta,\ (u,v) \in \mathcal{I}_X} F(\theta, u+v, u-v),$$

$$\inf_{\theta \in \Theta,\ p,q \in S_X} F(\theta, p, q) = \inf_{\theta \in \Theta,\ (u,v) \in \mathcal{I}_X} F(\theta, u+v, u-v).$$

The same is true after imposing any predicate on the original variables and its pullback under H.

Proof. If $p, q \in S_X$ and $(u,v) = H(p,q)$, then $u + v = p$ and $u - v = q$, so $\|u+v\| = \|u-v\| = 1$. Thus $(u,v) \in \mathcal{I}_X$. Conversely, if $(u,v) \in \mathcal{I}_X$, then $p = u+v$ and $q = u-v$ belong to S_X, and the two displayed linear maps undo each other. Every assertion about functions, extrema, parameters, and predicates follows by reindexing along this bijection. ☐

Corollary 1 (Finite sphere and ball constants)**.** *Every functional of finitely many unit vectors has an exact representation using finitely many isosceles-orthogonal pairs, by applying Theorem 1 to successive pairs of variables (and adding one ignored dummy sphere variable if their number is odd).*

Moreover, for every extended-real $G : B_X^2 \to [-\infty, \infty]$,

$$\sup_{x,y \in B_X} G(x,y) = \sup_{\substack{(u,v)\in \mathcal{I}_X \\ r,s\in[0,1]}} G\big(r(u+v), s(u-v)\big),$$

and similarly for infima and constraints.

Proof. Only the ball formula needs comment. Write a nonzero $x \in B_X$ as $x = rp$, where $r = \|x\| \in (0,1]$ and $p \in S_X$; if $x = 0$, take $r = 0$ and any $p \in S_X$. Do the same for $y = sq$, and apply Theorem 1 to (p,q). The converse inclusion is immediate. □

Thus, in the exact reparameterization sense, Problems 5.8 and 5.10 have the maximal answer: all finite-point sphere constants transfer, and all finite-point ball constants transfer after retaining their radial variables.

4 Arbitrarily many parameters

For an integer $N \geq 1$ and a nonzero matrix $A = (a_{\ell j}) \in \mathbb{R}^{N\times 2}$, define

$$\mathcal{L}_A(X) := \frac{1}{\|A\|_F^2} \sup_{p,q\in S_X} \sum_{\ell=1}^{N} \|a_{\ell 1}p + a_{\ell 2}q\|^2. \tag{1}$$

Up to common scaling, this is a family with $2N-1$ scalar parameters.

Theorem 2 (Matrix-coefficient L_{YJ} family)**.** *For every nonzero* $A \in \mathbb{R}^{N\times 2}$,

$$\mathcal{L}_A(X) = \frac{1}{\|A\|_F^2} \sup_{\substack{u\perp_I v \\ (u,v)\neq(0,0)}} \frac{\sum_{\ell=1}^{N} \|(a_{\ell 1}+a_{\ell 2})u + (a_{\ell 1}-a_{\ell 2})v\|^2}{\|u+v\|^2}. \tag{2}$$

For

$$A = \begin{pmatrix} \tau & \upsilon \\ \upsilon & -\tau \end{pmatrix},$$

the constant in (1) *is exactly* $L'_{\mathrm{YJ}}(\tau, \upsilon, X)$ *from the source.*

Proof. For a nonzero isosceles pair put $r = \|u+v\| = \|u-v\| > 0$ and $p = (u+v)/r$, $q = (u-v)/r$. Then

$$a_{\ell 1}p + a_{\ell 2}q = \frac{(a_{\ell 1}+a_{\ell 2})u + (a_{\ell 1}-a_{\ell 2})v}{r}.$$

Theorem 1 gives (2). For the displayed two-row matrix, $\|A\|_F^2 = 2(\tau^2+\upsilon^2)$, and the numerator in (1) is $\|\tau p + \upsilon q\|^2 + \|\upsilon p - \tau q\|^2$, which is the defining formula for L'_{YJ}. □

Since N is arbitrary, Theorem 2 answers Problem 5.9 with a single family containing arbitrarily many independent coefficients.

5 Which other orthogonalities transfer?

Let $\perp_R$ be any relation between a nonzero first vector and a second vector in X.

Definition 1. *The relation $\perp_R$ is* rectifiable *if, for every $x \neq 0$ and every $z \in X$, there are $\beta \in \mathbb{R}$ and $y \in X$ such that*

$$z = \beta x + y, \qquad x \perp_R y.$$

It is sphere-rectifiable *if the same property holds whenever $x = (p+q)/2 \neq 0$ and $z = (p-q)/2$ for $p, q \in S_X$.*

Theorem 3 (Rectifiability characterization)**.** *The following are equivalent for a relation $\perp_R$.*

1. *The relation is rectifiable.*
2. *The coordinate map*
$$(x, y, \beta) \longmapsto (x, \beta x + y), \qquad x \perp_R y,$$
is onto $(X \setminus \{0\}) \times X$.
3. *For every bounded function $\Phi : (X \setminus \{0\}) \times X \to \mathbb{R}$,*
$$\sup_{x \neq 0, z \in X} \Phi(x, z) = \sup_{\substack{x \perp_R y \\ x \neq 0,\ \beta \in \mathbb{R}}} \Phi(x, \beta x + y),$$
and likewise for infima.

The analogous universal transfer on S_X^2 holds if and only if the relation is sphere-rectifiable, with antipodal pairs $(p, -p)$ included as a separate boundary component. Every rectifiable relation is sphere-rectifiable.

Proof. The first two assertions are the same surjectivity statement. Surjectivity implies the extrema identities by substitution. Conversely, if a pair (x_0, z_0) is missed, take Φ to be its indicator: the left supremum is 1 and the right one is 0. The sphere statement applies the same argument to

$$p = \frac{(1+\beta)x + y}{r}, \qquad q = \frac{(1-\beta)x - y}{r},$$

where the two numerator norms have the same positive value r. A non-antipodal pair has $x = (p+q)/2 \neq 0$ and is reached exactly when its half-difference can be rectified. Antipodal pairs have $x = 0$ and form the stated boundary component. □

This is a necessary-and-sufficient formal answer to Problem 5.11. The current source proves that Birkhoff–James orthogonality is rectifiable by a norming functional. The next lemma supplies a different named example.

Lemma 1 (Pythagorean orthogonality is rectifiable)**.** *Define $x \perp_P y$ by*

$$\|x - y\|^2 = \|x\|^2 + \|y\|^2.$$

Then $\perp_P$ is rectifiable on every real normed space.

Proof. Fix $x \neq 0$ and $z \in X$, and put $q(t) = \|z - tx\|^2$. The function q is continuous, convex, and satisfies

$$\frac{q(t)}{t^2} \longrightarrow \|x\|^2 \qquad (|t| \to \infty).$$

Its forward difference $d(t) = q(t+1) - q(t)$ is continuous and nondecreasing. For $t > 0$, convexity gives

$$d(t) \geq \frac{q(t) - q(0)}{t} \longrightarrow +\infty.$$

For $t < -1$, comparison of the secant slopes on $[t, t+1]$ and $[t+1, 0]$ gives

$$d(t) \leq \frac{q(0) - q(t+1)}{-t-1} \longrightarrow -\infty.$$

Hence there is $\beta \in \mathbb{R}$ with $d(\beta) = \|x\|^2$. Set $y = z - \beta x$. Since

$$\|y\|^2 = q(\beta), \qquad \|x - y\|^2 = q(\beta + 1),$$

we obtain $\|x - y\|^2 = \|x\|^2 + \|y\|^2$, as required. $\square$

Theorems 3 and Lemma 1 now produce exact Pythagorean-orthogonal representations for every two-vector full-space constant, and—with the equal-radius endpoint condition—for every sphere constant.

6 Operator constants and moduli

Corollary 2 (Universal operator transfer)**.** *Let $T : X \to Y$ be a bounded linear operator, let Θ be any parameter set, and let $\Psi : \Theta \times Y^2 \to [-\infty, \infty]$. Then*

$$\sup_{\theta \in \Theta,\ p,q \in S_X} \Psi(\theta, Tp, Tq) = \sup_{\theta \in \Theta,\ (u,v) \in \mathcal{I}_X} \Psi\big(\theta, T(u+v), T(u-v)\big),$$

with identical statements for infima and arbitrary constraints. Pairwise application gives the same result for every finite-point operator constant or modulus, and Corollary 1 gives unit-ball versions.

Proof. Apply Theorem 1 to $F(\theta, p, q) = \Psi(\theta, Tp, Tq)$. Linearity gives the displayed form. Products of the bijection handle more variables, and predicates pull back exactly. $\square$

For example, (1) has the operator analogue

$$\mathcal{L}_A(T) := \frac{1}{\|A\|_F^2} \sup_{p,q \in S_X} \sum_{\ell=1}^{N} \|T(a_{\ell 1} p + a_{\ell 2} q)\|_Y^2,$$

and Theorem 2 remains valid after applying T to every vector in its numerator. Constraints appearing in operator moduli of convexity or smoothness transfer just as exactly. This answers Problem 5.12 at the level of every finite-configuration operator invariant.

7 Verification, limitations, and novelty

The proof uses only explicit bijections, surjectivity, and one-variable convexity. The main checks are: the zero half-sum occurs exactly at antipodal sphere pairs and is retained as a boundary component; a nonzero isosceles pair has $\|u + v\| > 0$; the matrix normalization recovers the source denominator; and the Pythagorean forward difference has the correct two infinite limits. No computational or unproved dependency remains.

The result gives a full answer only under the precise interpretation that an "equivalent expression" means an exact reparameterization, with auxiliary scalar variables allowed. The source does not define a narrower notion such as formula complexity, naturality, or usefulness. If one of those was intended, the theorem is instead a complete structural baseline: it says existence is automatic and identifies what additional criterion must be specified for a nontrivial classification.

The run indexes and a bounded arXiv/web search through 17 August 2026 found many particular orthogonal constants but no universal finite-point theorem, arbitrary matrix family, rectifiability characterization, or Pythagorean transfer. Searches used the exact title and problems, L'_{YJ} with matrix/multiple parameters, unit-sphere transfer, other orthogonalities, and operator versions; close papers included arXiv:2111.08392, 2504.00826, 2509.23319, 2602.13974, 2606.01064, and 2606.01068. Novelty is plausible, not certified.

Human review should focus on whether the literal formalization matches the authors' intended breadth, the necessity direction of Theorem 3, and the secant-slope limits in Lemma 1. Subject to those checks and a specialist search, promote as a full structural resolution of Problems 5.8–5.12.

References

[1] Q. Liu, Y. Wang, and M. Bao, *How orthogonality influences geometric constants*, arXiv:2507.17122v5, Problems 5.8–5.12 (2025–2026).